PREDICTIVE MODELLING – LINEAR REGRESSION

SHOWTIME – OTT SERVICES

APRIL 20, 2025

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Problem Statement

Business Context

An over-the-top (OTT) media service is a media service offered directly to viewers via the internet. The term is most synonymous with subscription-based video-on-demand services that offer access to film and television content, including existing series acquired from other producers, as well as original content produced specifically for the service. They are typically accessed via websites on personal computers, apps on smartphones and tablets, or televisions with integrated Smart TV platforms.

Presently, OTT services are at a relatively nascent stage and are widely accepted as a trending technology across the globe. With the increasing change in customers' social behavior, which is shifting from traditional subscriptions to broadcasting services and OTT on-demand video and music subscriptions every year, OTT streaming is expected to grow at a very fast pace. The global OTT market size was valued at $121.61 billion in 2019 and is projected to reach $1,039.03 billion by 2027, growing at a CAGR of 29.4% from 2020 to 2027. The shift from television to OTT services for entertainment is driven by benefits such as on-demand services, ease of access, and access to better networks and digital connectivity.

With the outbreak of COVID19, OTT services are striving to meet the growing entertainment appetite of viewers, with some platforms already experiencing a 46% increase in consumption and subscriber count as viewers seek fresh content. With innovations and advanced transformations, which will enable the customers to access everything they want in a single space, OTT platforms across the world are expected to increasingly attract subscribers on a concurrent basis.

Objective

ShowTime is an OTT service provider and offers a wide variety of content (movies, web shows, etc.) for its users. They want to determine the driver variables for first-day content viewership so that they can take necessary measures to improve the viewership of the content on their platform. Some of the reasons for the decline in viewership of content would be the decline in the number of people coming to the platform, decreased marketing spend, content timing clashes, weekends and holidays, etc. They have hired you as a Data Scientist, shared the data of the current content in their platform, and asked you to analyze the data and come up with a linear regression model to determine the driving factors for first-day viewership.

Data Description

1. visitors: Average number of visitors, in millions, to the platform in the past week
2. ad_impressions: Number of ad impressions, in millions, across all ad campaigns for the content (running and completed)
3. major_sports_event: Any major sports event on the day
4. genre: Genre of the content
5. dayofweek: Day of the release of the content
6. season: Season of the release of the content
7. views_trailer: Number of views, in millions, of the content trailer
8. views_content: Number of first-day views, in millions, of the content

Data Overview

The initial steps to get an overview of any dataset is to:

- observe the first few rows of the dataset, to check whether the dataset has been loaded properly or not
- get information about the number of rows and columns in the dataset
- find out the data types of the columns to ensure that data is stored in the preferred format and the value of each property is as expected.
- check the statistical summary of the dataset to get an overview of the numerical columns of the data

Displaying the first 5 rows of dataset

	visitors	ad_impressions	major_sports_event	genre	dayofweek	season	views_trailer	views_content
0	1.67	1113.81	0	Horror	Wednesday	Spring	56.70	0.51
1	1.46	1498.41	1	Thriller	Friday	Fall	52.69	0.32
2	1.47	1079.19	1	Thriller	Wednesday	Fall	48.74	0.39
3	1.85	1342.77	1	Sci-Fi	Friday	Fall	49.81	0.44
4	1.46	1498.41	0	Sci-Fi	Sunday	Winter	55.83	0.46

Displaying the shape of the dataset

```
Data has 1000 row and 8 columns

(1000, 8)
```

Displaying the Datatypes of the Columns

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 8 columns):
 #   Column              Non-Null Count  Dtype
---  ------              --------------  -----
 0   visitors            1000 non-null   float64
 1   ad_impressions      1000 non-null   float64
 2   major_sports_event  1000 non-null   int64
 3   genre               1000 non-null   object
 4   dayofweek           1000 non-null   object
 5   season              1000 non-null   object
 6   views_trailer       1000 non-null   float64
 7   views_content       1000 non-null   float64
dtypes: float64(4), int64(1), object(3)
memory usage: 62.6+ KB
```

Displaying the statistical Summary

	visitors	ad_impressions	major_sports_event	views_trailer	views_content
count	1000.000000	1000.000000	1000.000000	1000.00000	1000.000000
mean	1.704290	1434.712290	0.400000	66.91559	0.473400
std	0.231973	289.534834	0.490143	35.00108	0.105914
min	1.250000	1010.870000	0.000000	30.08000	0.220000
25%	1.550000	1210.330000	0.000000	50.94750	0.400000
50%	1.700000	1383.580000	0.000000	53.96000	0.450000
75%	1.830000	1623.670000	1.000000	57.75500	0.520000
max	2.340000	2424.200000	1.000000	199.92000	0.890000

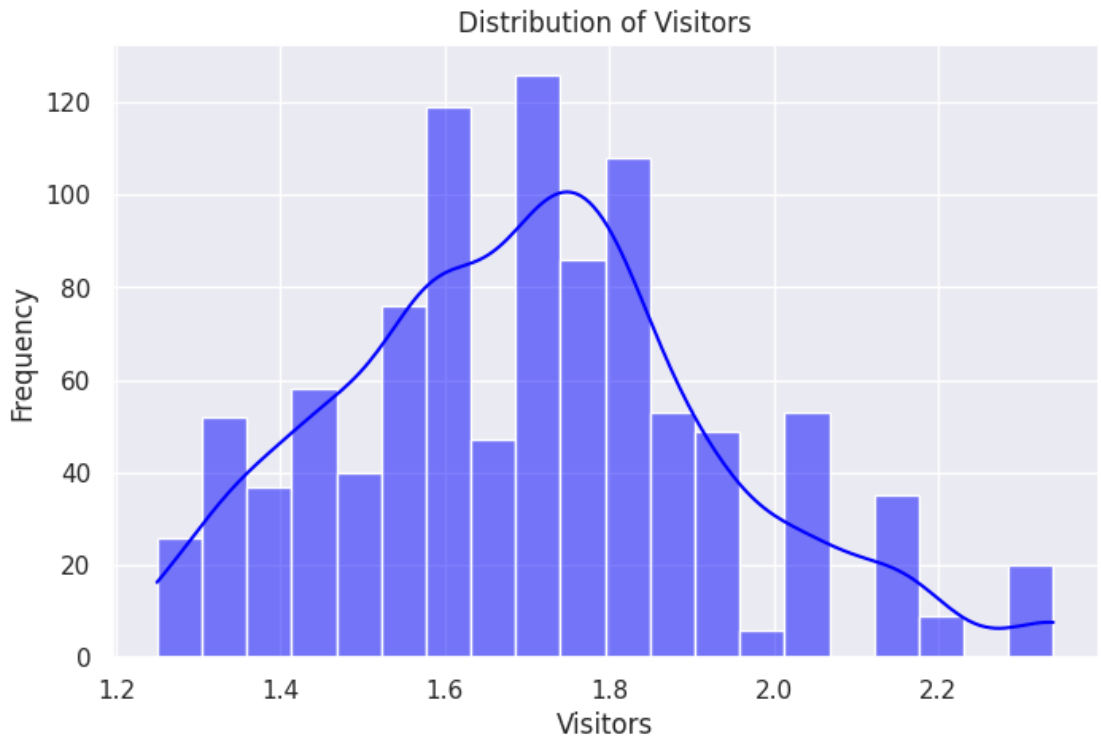
Checking the missing values

```
visitors               0
ad_impressions         0
major_sports_event     0
genre                  0
dayofweek              0
season                 0
views_trailer          0
views_content          0
dtype: int64
```

Exploratory Data Analysis

Univariate Analysis

Visitors

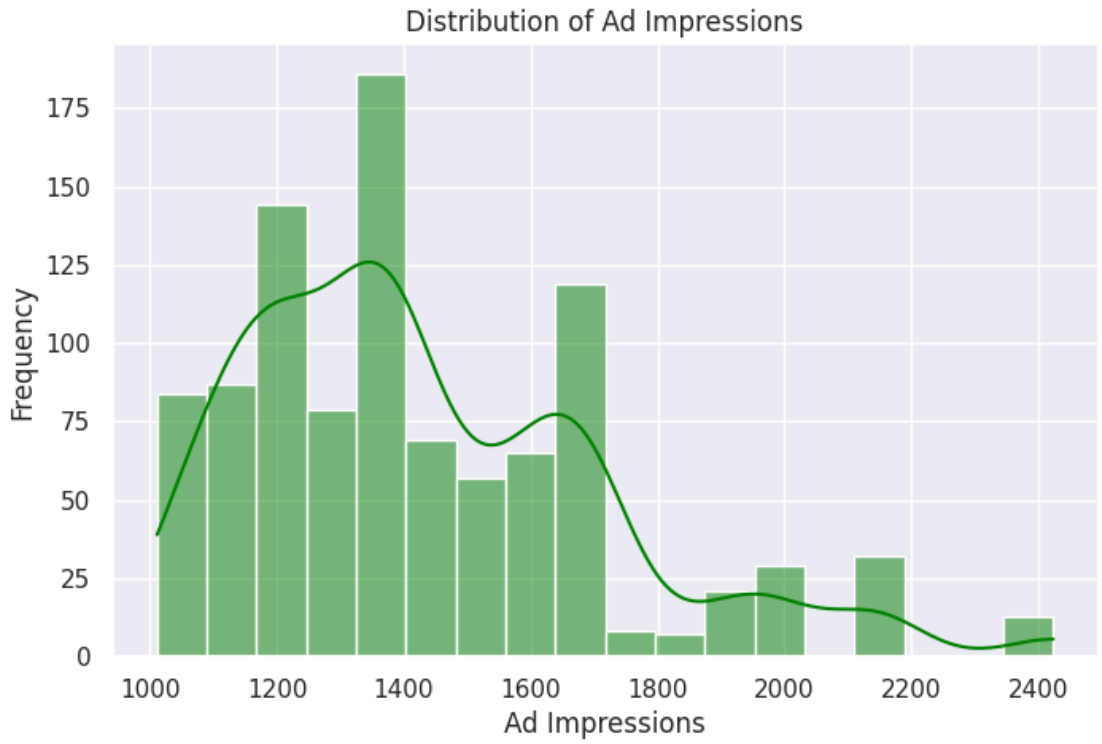


Observation

Shape: Approximately normal distribution with a slight right skew Range: Primarily between 1.2 and 2.2 (likely in millions) Peak: Highest frequency around 1.7-1.8 Key Observations:

Most content receives between 1.6-1.8 million visitors There's a gradual decline in frequency for both very low (<1.4) and very high (>2.0) visitor counts The relatively symmetrical distribution suggests stable platform traffic patterns

Ad Impressions

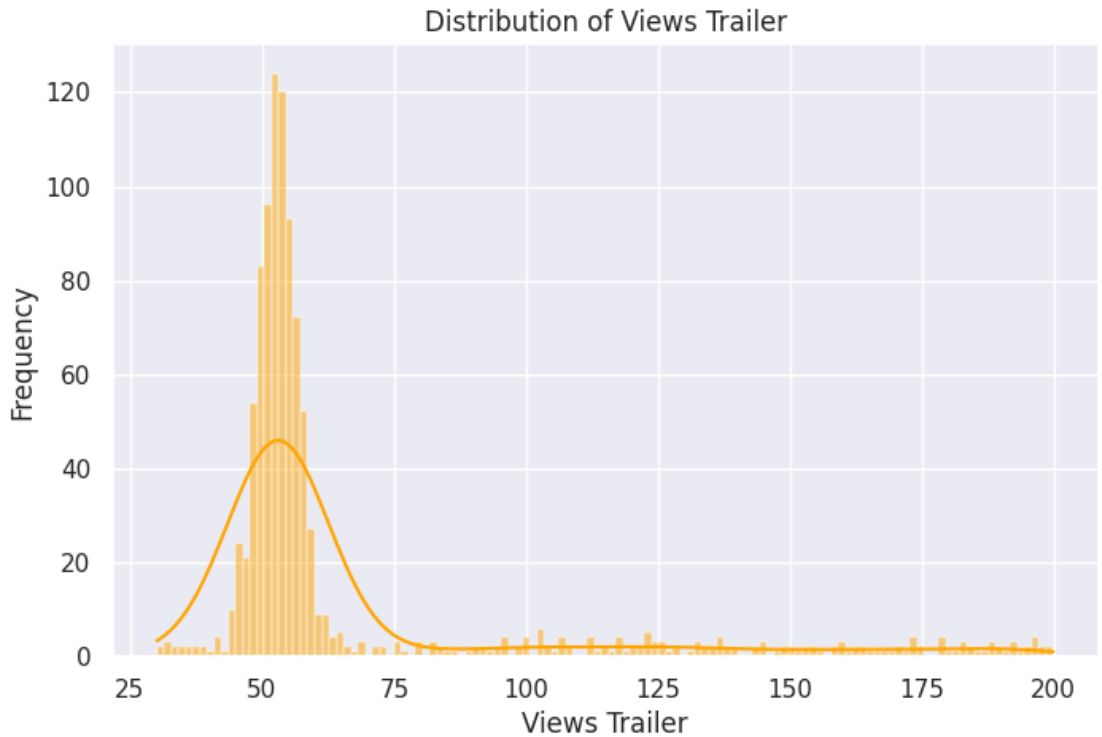


Observation

Shape: Bimodal distribution (two peaks) Range: 1000-2400 impressions Primary Peak: Around 1300-1400 impressions Secondary Peak: Around 1600-1700 impressions Key Observations:

The bimodal nature suggests two distinct advertising strategies or campaigns Majority of content receives between 1200-1800 ad impressions Long right tail indicates some content getting significantly higher advertising exposure This could indicate targeted marketing for certain content types

Views Trailers

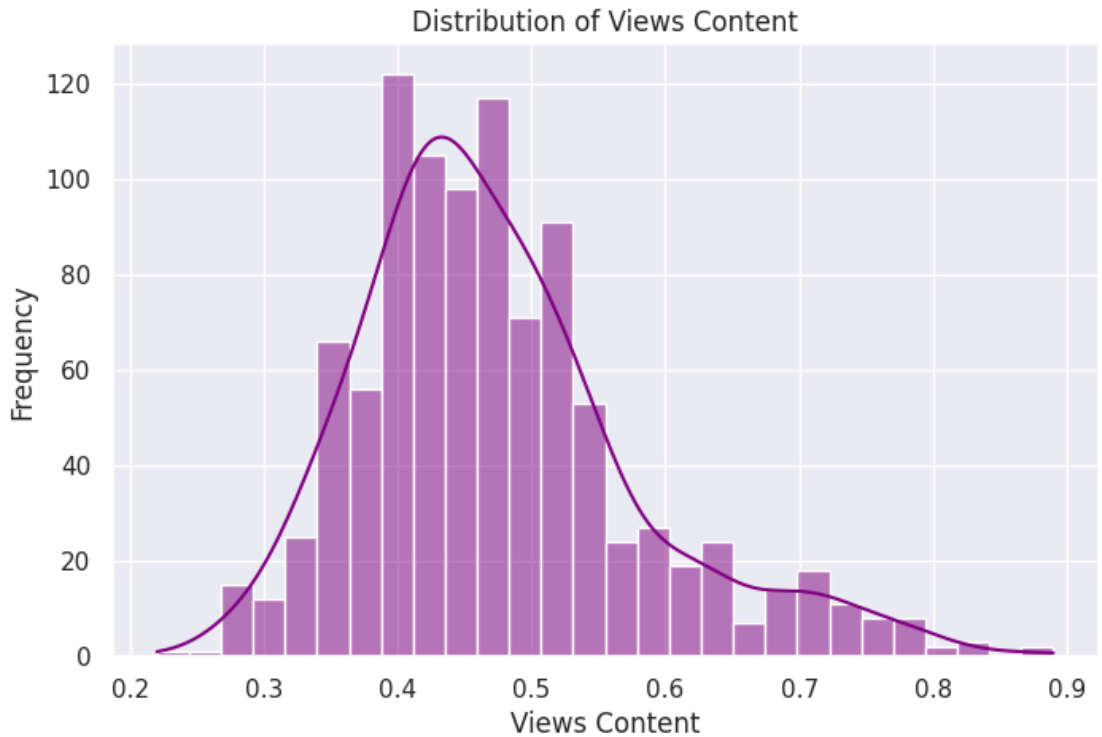


Observation

Shape: Highly right-skewed with a sharp peak Range: 25-200 views Peak: Concentrated around 50 views Key Observations:

Most trailers receive around 50 views Very few trailers receive more than 75 views The sharp peak suggests consistency in trailer viewing behavior Long right tail indicates a few highly successful trailer campaigns

Views Content

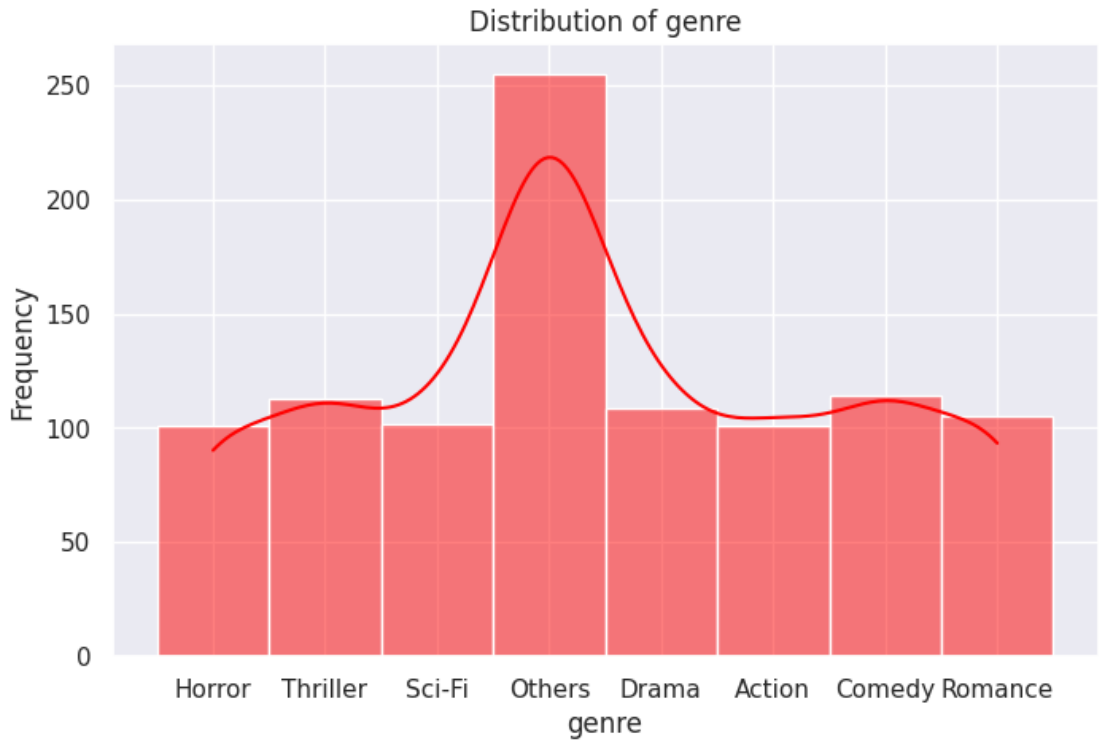


Observation

Shape: Slightly right-skewed with a clear central tendency Range: 0.2-0.9 (likely in millions) Peak: Around 0.4-0.5 Key Observations:

Most content receives between 0.4-0.5 million views More symmetric than trailer views, suggesting more predictable content viewing patterns Gradual decline in higher view counts indicates natural viewing limits

Genres



Observations

Genre Distribution Pattern:

The distribution is multimodal, with several peaks across different genres "Others" category shows the highest frequency with around 250 entries Most other genres have frequencies between 100-125 entries

Genre-specific Frequencies:

"Others" category is significantly overrepresented compared to specific genres Horror, Thriller, Sci-Fi show similar frequencies (around 100 entries each) Comedy and Romance also show comparable frequencies to each other Drama and Action have slightly lower frequencies than other specific genres

Category Groupings:

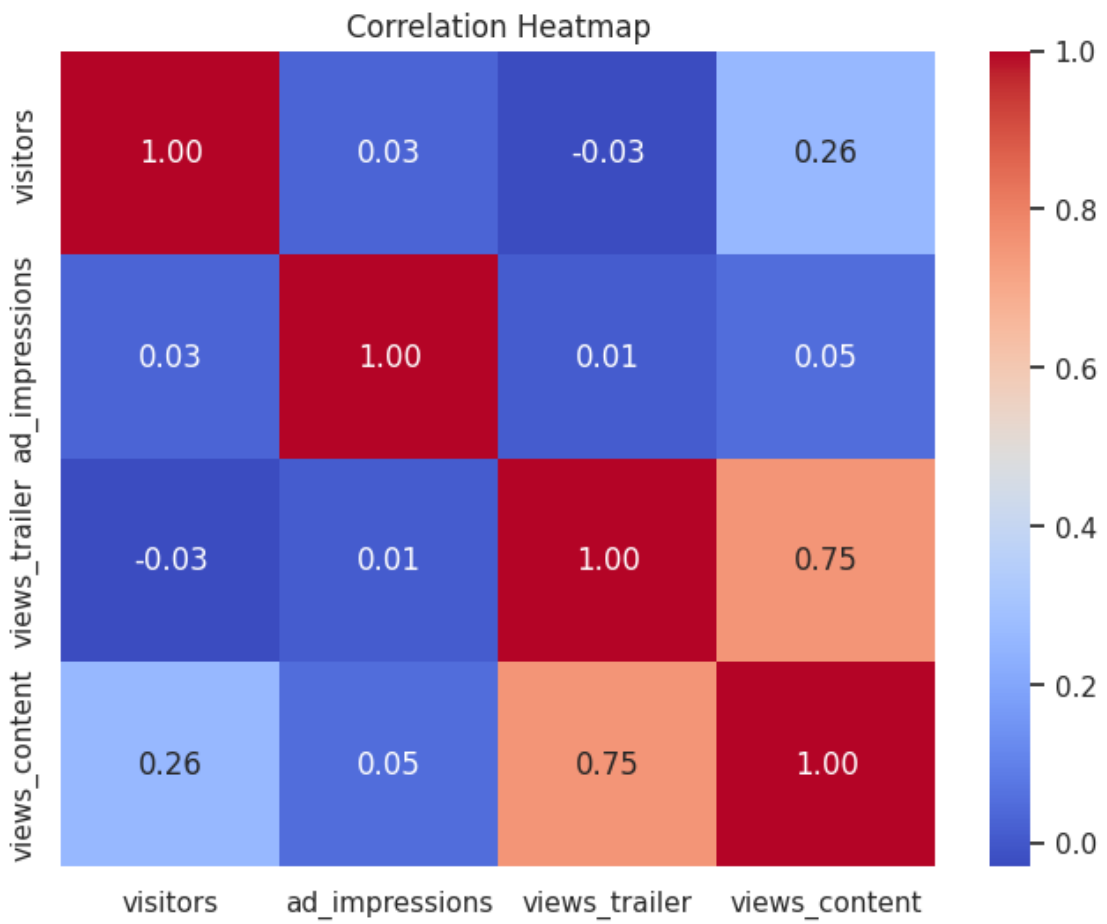
Thriller and Sci-Fi genres show similar frequencies, suggesting comparable popularity Comedy and Romance have nearly identical frequencies, possibly indicating viewer overlap Horror maintains a moderate presence in the collection

Balance Analysis:

Excluding the "Others" category, there's relatively even distribution across genres The specific genres show a balanced representation in the content library No specific genre (excluding "Others") dominates or is significantly underrepresented

Bivariate Analysis

Correlation Heatmap



Observation

1. **Strong Positive Correlation**:

The strongest correlation (0.75) exists between views_trailer and views_content This suggests that people who watch trailers are likely to watch the actual content The relationship makes intuitive sense as interest in trailers often translates to content viewing

1. **Weak Correlations**:

Visitors and ad_impressions show almost no correlation (0.03) Ad_impressions have very weak correlations with all other metrics:

Only 0.01 with views_trailer 0.05 with views_content

This suggests ad impressions operate somewhat independently of other metrics

1. **Moderate Correlations**:

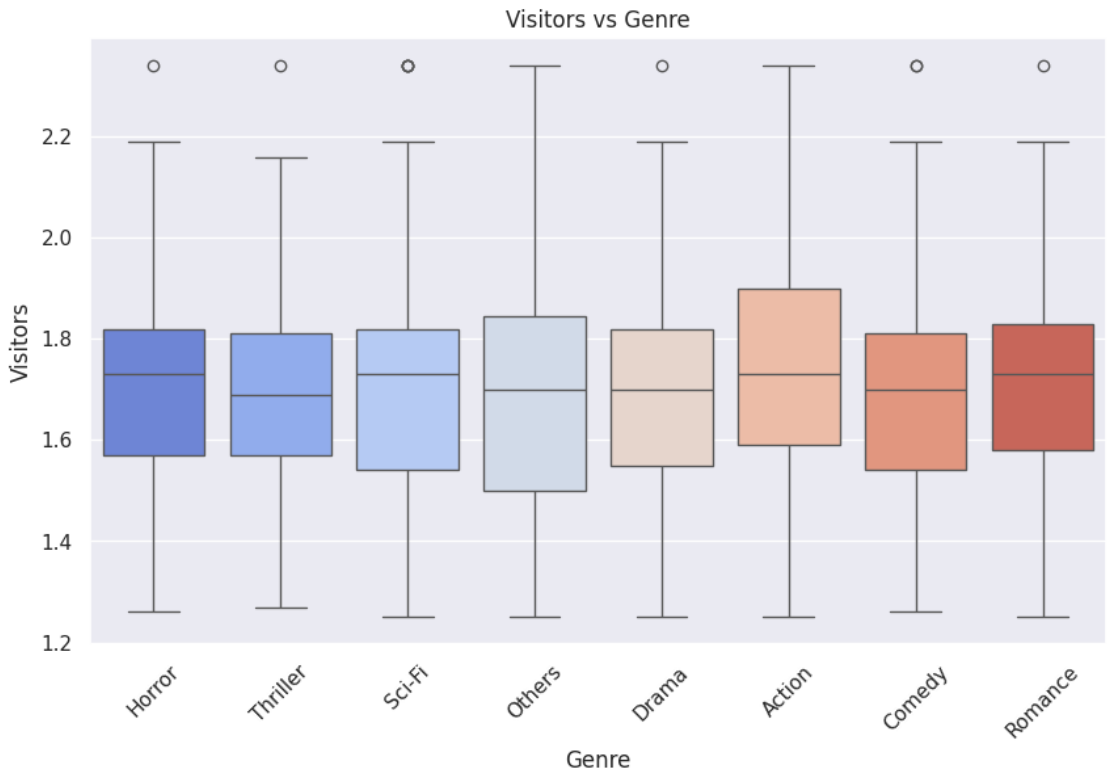
Visitors have a weak positive correlation (0.26) with views_content This indicates that higher visitor numbers somewhat translate to more content views, but the relationship isn't very strong Visitors have a slight negative correlation (-0.03) with views_trailer

1. **Pattern Insights**:

The effectiveness of ads seems questionable given their weak correlations with viewing metrics Trailer views are a much better predictor of content viewing than visitor numbers or ad impressions

1. Visitors vs All Other Variables

Visitors vs Genre



Observation

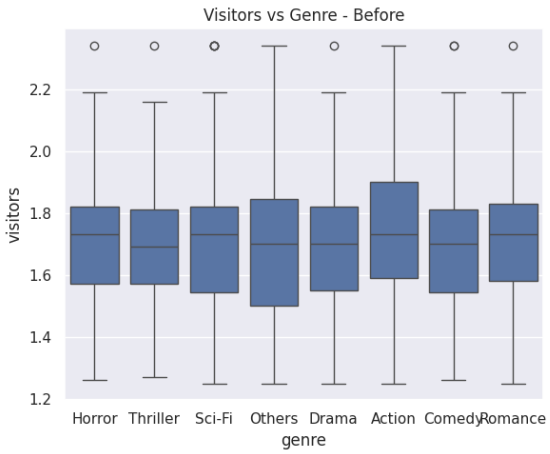
Distribution Patterns:

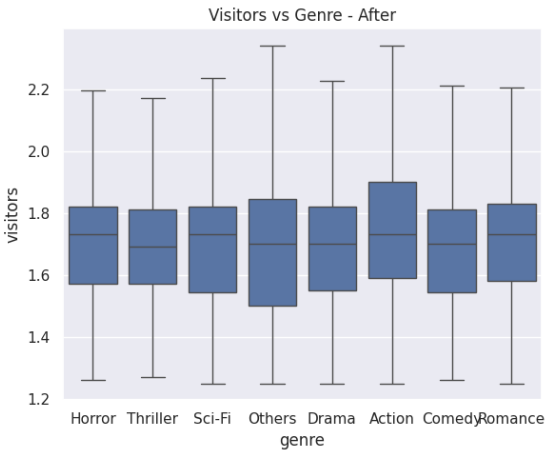
Action shows highest median visitors (~1.75) Romance and Comedy follow closely Horror, Thriller, and Sci-Fi show similar visitor patterns 'Others' category shows highest variability

Key Insights:

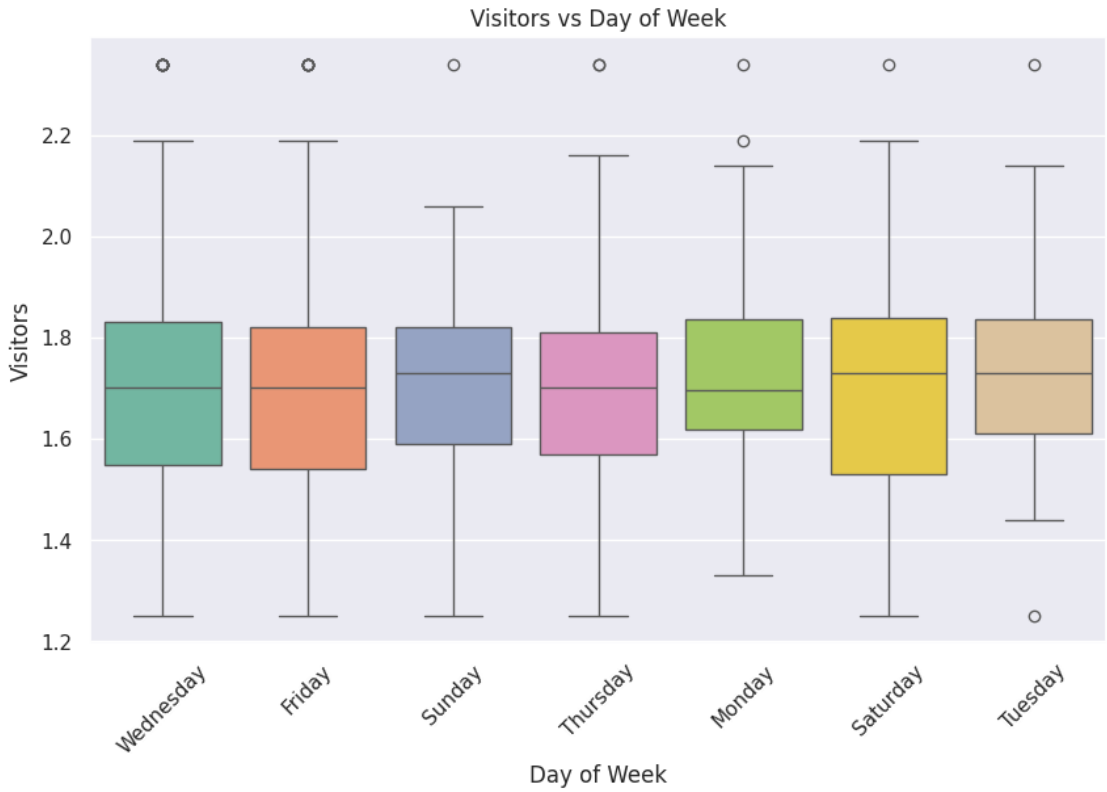
Action content drives more platform traffic Traditional genres (Horror, Thriller) show consistent but lower visitor numbers All genres maintain baseline visitors between 1.4-1.8 million

Outlier Treatment

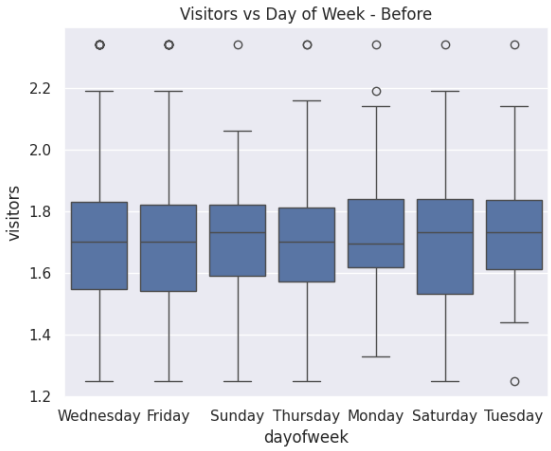


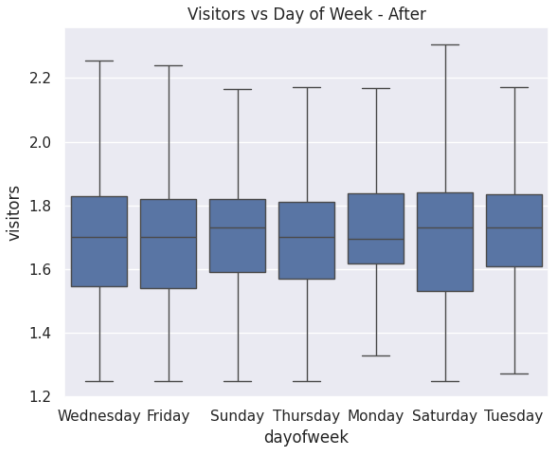


 # Visitors vs Day of Week



Outlier Treatment





Observation

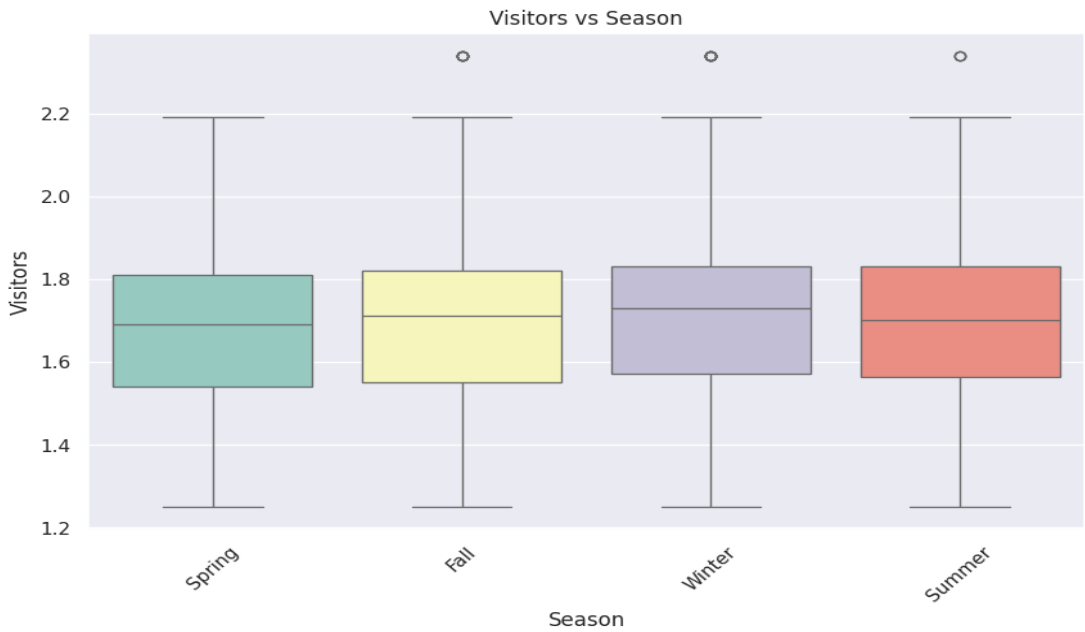
Temporal Patterns:

Saturday shows highest median visitors Weekdays (Tuesday-Thursday) show relatively consistent patterns Weekend days (Friday-Sunday) show slightly higher visitor numbers Monday shows moderate visitor numbers

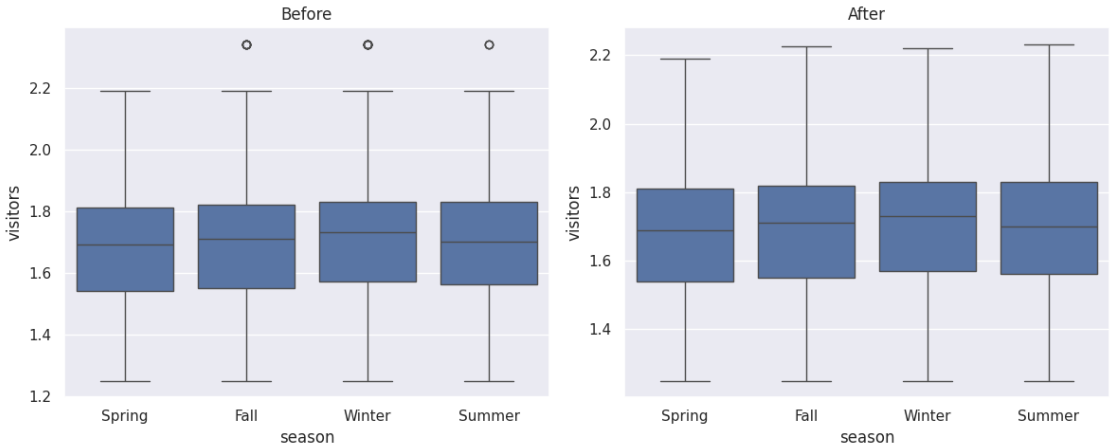
Before vs After Comparison:

More consistent patterns in the "After" period Reduced outliers in recent data Slightly higher median values across most days Weekend preference remains consistent

Visitors vs Season



Outlier Treatment



Observation

From the original seasonal distribution (Image 1):

Seasonal Patterns:

All seasons show similar median visitor numbers (around 1.7) The interquartile ranges (box sizes) are relatively consistent across seasons Each season has some outliers at higher visitor numbers (around 2.2) The distribution appears fairly symmetric for all seasons

Looking at the Before/After outlier treatment (Image 2):

Changes After Treatment:

The extreme outliers at the top (around 2.3-2.4) have been removed The overall shape and central tendency of the distributions remain largely unchanged The whiskers (vertical lines extending from boxes) are now more uniform across seasons The boxes (interquartile ranges) maintain their original positions

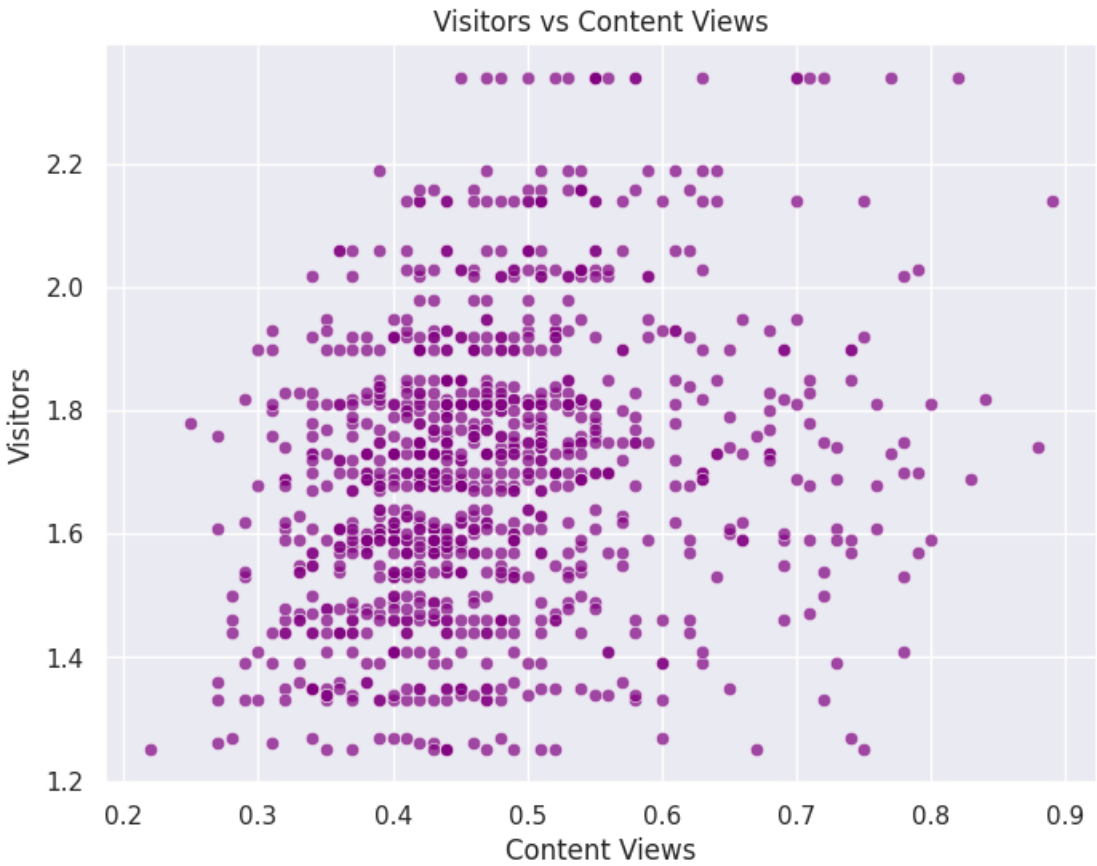
Impact of Treatment:

The outlier treatment has preserved the underlying seasonal patterns The data is now more consistent across seasons The treatment appears to have been conservative, maintaining the natural variation while removing only extreme values Winter and Summer show slightly higher medians after treatment, but the difference is minimal

Overall, this suggests that:

Visitor numbers are relatively stable across seasons The outlier treatment successfully removed extreme values while preserving the natural seasonal patterns There's no strong seasonal effect on visitor numbers The data cleaning has made the distributions more comparable across seasons without losing important trends

Visitors vs Views Content



Observation

Relationship Pattern:

There's a positive correlation between visitors and content views The relationship appears to be somewhat linear, but with considerable scatter The density of points is highest in the middle range (around 0.4-0.6 content views)

Distribution Features:

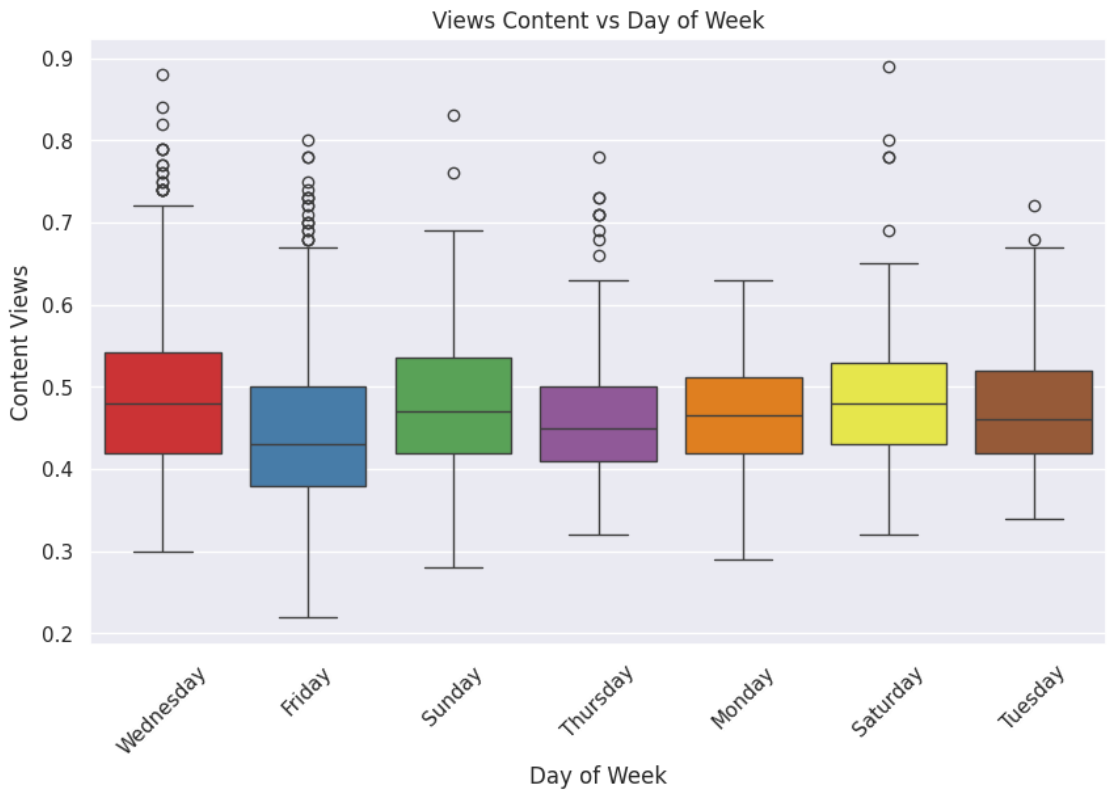
Most data points cluster between 1.4 and 2.0 visitors Content views mostly fall between 0.3 and 0.7 There are some outlier points, particularly in the upper right quadrant The scatter becomes more dispersed as content views increase

Notable Insights:

Higher content views generally correspond to higher visitor numbers There's significant variation in visitor numbers for any given level of content views The relationship seems to have more variability at higher content view levels

2. Views Content vs Other Variables

Views Content vs Day of Week



Observation

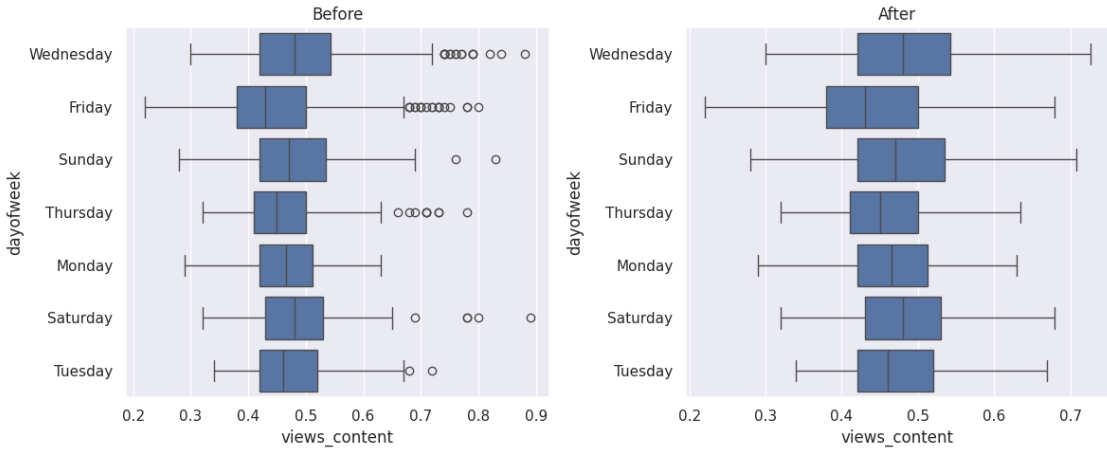
Daily Patterns:

Wednesday shows the highest median content views and largest interquartile range Friday appears to have the lowest median content views The distribution is relatively consistent across other days All days show several outliers (dots) above the whiskers, indicating some unusually high viewing periods

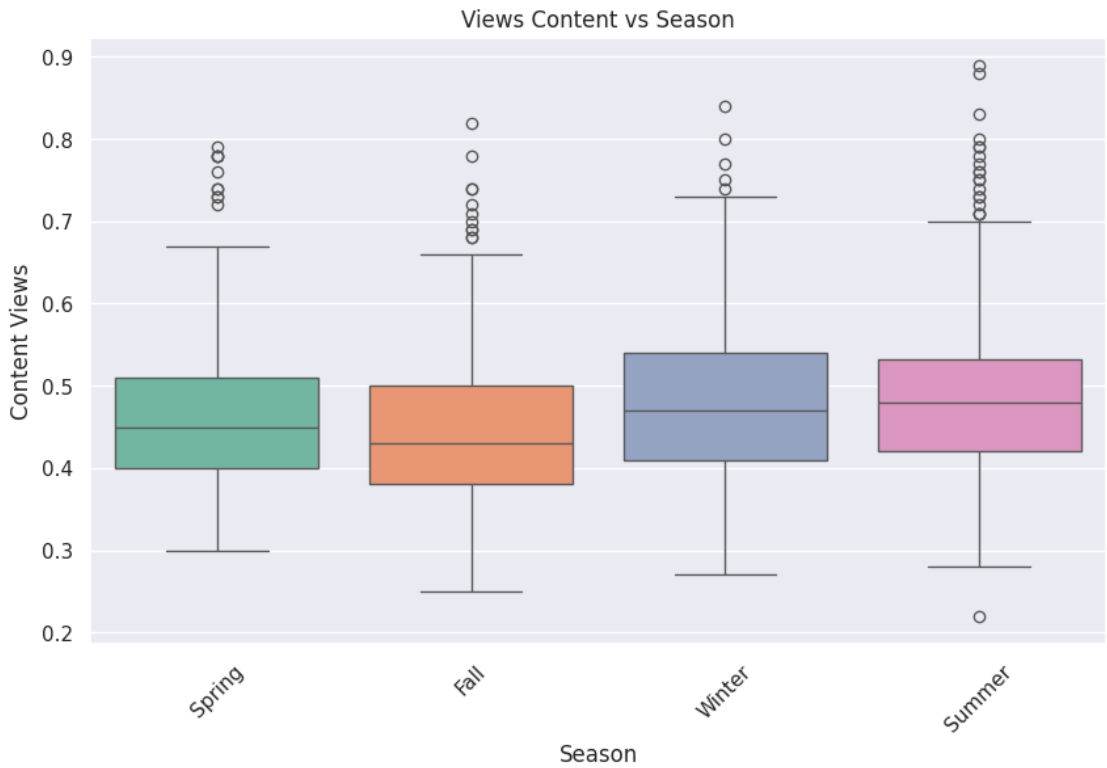
Distribution Characteristics:

Each day shows a similar interquartile range (box size), suggesting consistent variability The medians (horizontal lines in boxes) are fairly stable across days, mostly between 0.4 and 0.5 There's slight positive skew across all days, shown by the higher position of outliers

Outlier Treatment



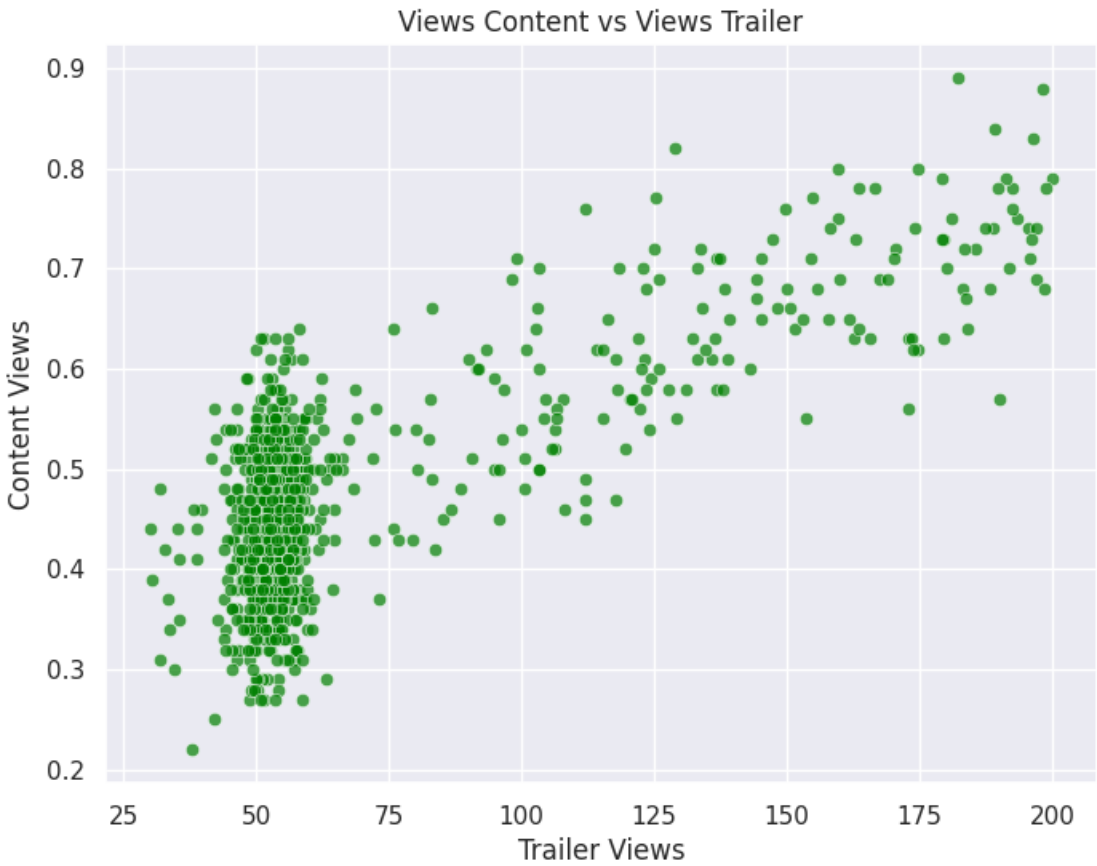
Views Content vs Season



Observation

The median content views are fairly consistent across seasons, hovering around 0.45-0.5 Winter appears to have a slightly higher median and larger spread There are several outliers in all seasons, particularly in Summer, with content views reaching up to 0.8-0.9 Summer and Winter show slightly more variability in their distributions compared to Spring and Fall

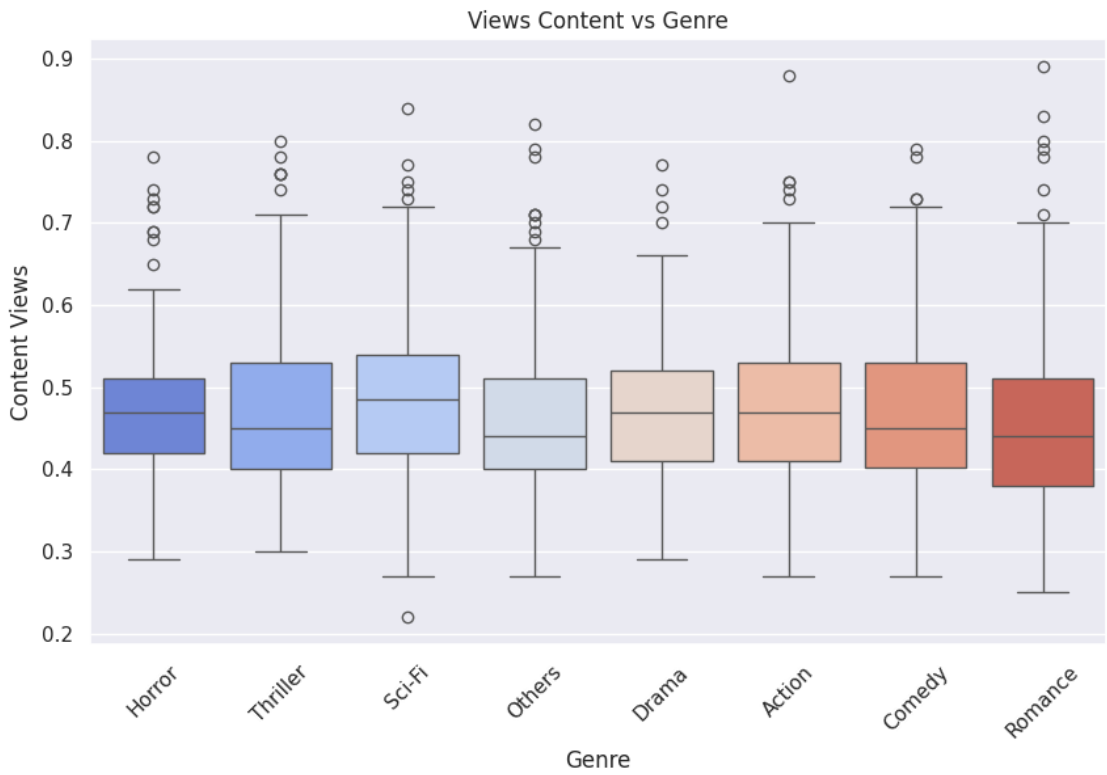
Views Content vs Views Trailer



Observation

There's a clear positive correlation between trailer views and content views The relationship appears to be non-linear, with the correlation becoming stronger as trailer views increase There's a dense cluster of data points around 50 trailer views, suggesting this might be a common baseline The spread of content views increases with higher trailer views, indicating more variability in content performance for videos with popular trailers The maximum content views (around 0.9) are associated with higher trailer views (150-200 range)

Views Content vs Genre

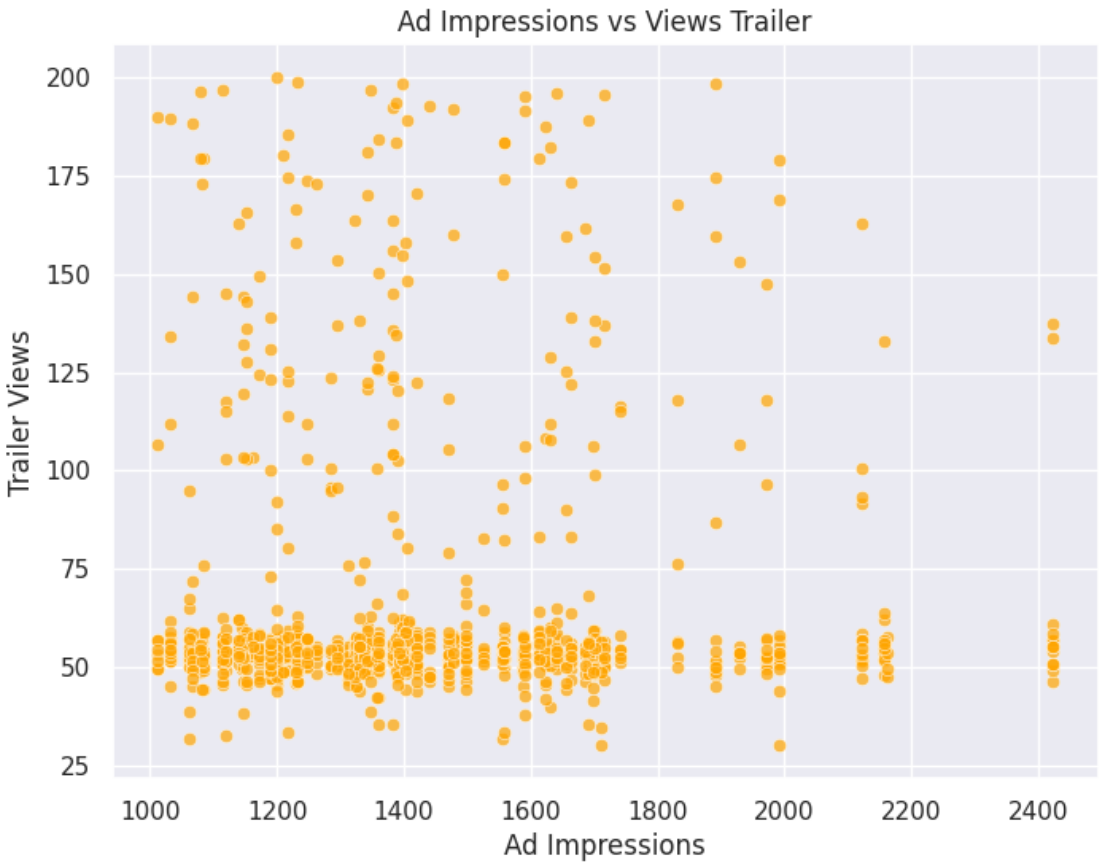


Observation

The median content views are relatively consistent across genres, typically between 0.4-0.5 Romance and Comedy show slightly higher variability in their distributions Sci-Fi and Thriller genres have several high-performing outliers (reaching 0.8+ views) The "Others" category shows the most compact distribution, suggesting more consistent but moderate performance

3. Ad Impressions

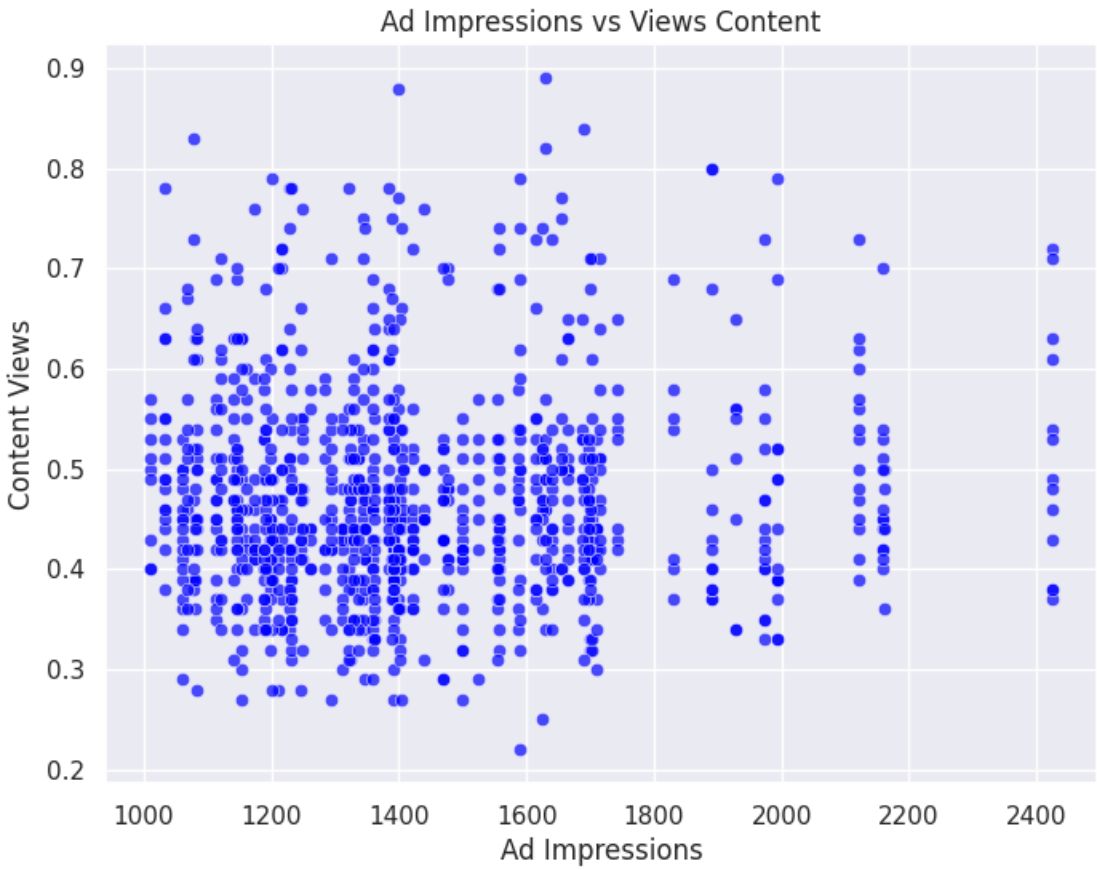
Ad Impressions vs Views Trailer



Observation

There's a clear concentration of trailer views around the 50-view mark across all ad impression levels There's a scattered distribution of higher trailer views (75-200) that doesn't show a strong correlation with ad impressions Ad impressions range primarily from 1000 to 2400, with the densest cluster between 1000-1800 The relationship doesn't appear to be strongly linear - increasing ad impressions doesn't necessarily lead to proportionally higher trailer views

```
Ad Impressions vs Views Content
```

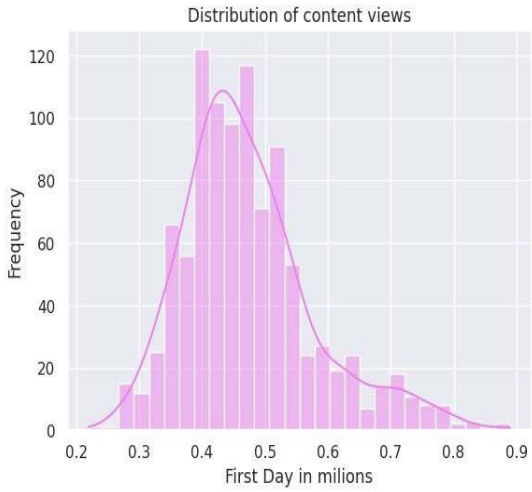


Observation

Content views are fairly dispersed between 0.3 and 0.8, regardless of ad impression count There's no clear correlation between ad impressions and content views The highest content views (0.8-0.9) appear across various ad impression levels The density of points is highest in the 0.4-0.6 content views range

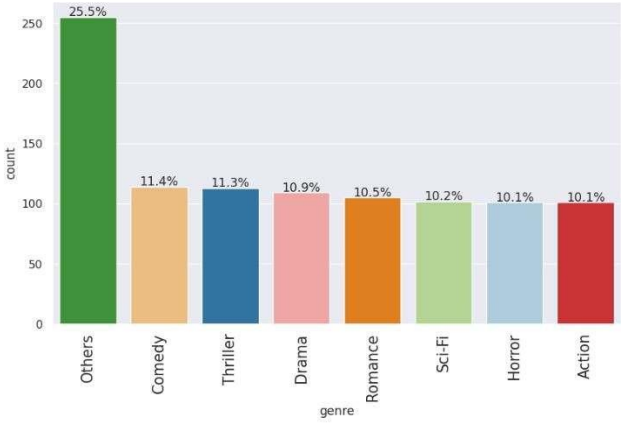
KEY QUESTIONS

1. What does the distribution of the content look like?



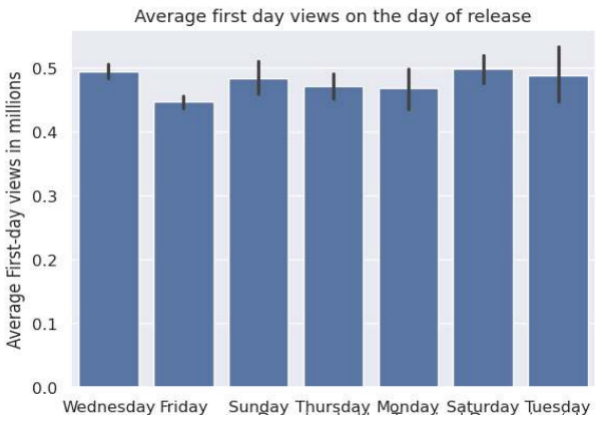
It has a normal distribution but slightly right skewed and the mean values lie between 4 to 5 million.

2.What does the distribution of genres look like?



The category others has the highest content release, followed by the genres comedy and thrillers

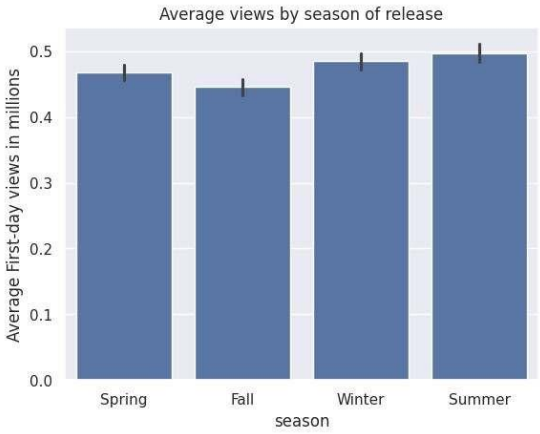
3.The day of the week on which content is released generally plays a key role in the viewership. How does the viewership vary with the day of release?



Wednesday and Saturday have the highest content release. Friday has the least viewership, while the other days have the

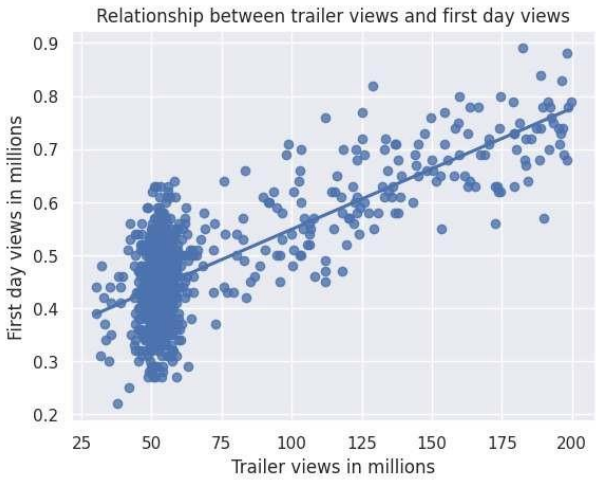
same average views during the content release.

4.How does the viewership vary with the season of release?



There is not much of an impact of season on the viewership and release of content. Summer has the highest viewership followed by Winter.

5. What is the correlation between trailer views and content views?



There is a positive relationship between Trailer views and , content views increase correspondingly.

DATA PROCESSING

1.Duplicate and Missing values

There are no duplicate or missing values.

	0
visitors	0
ad_impressions	0
major_sports_event	0
genre	0
dayofweek	0
season	0
views_trailer	0
views_content	0

dtype: int64

2. Feature Engineering

1. **Data Copy:**
   Making a copy of the input data to keep the original dataset unchanged.

2. **Categorical Encoding:**
   Converting the genre column into multiple binary columns (one-hot encoding) and also turn the day o fweek and season columns into numeric codes. This makes all categorical data usable for modeling.

3. **Cyclical Feature Creation:**
   To capture repeating patterns, transform the numeric day-of-week and season values using sine and cosine functions, allowing the model to recognize cycles in the data.

4. **Numerical Transformations:**
   Applying a logarithmic transformation to features like ad impressions, content views, and trailer views. This reduces the effect of extreme values and helps normalize their distributions.

5. **Interaction Features:**
   Creating new features by multiplying content views with both the day-of-week code and the genre code. These capture how the effect of content views might change depending on the day or genre.

6. **Derived Features:**
   Calculate engagement per visitor (content views divided by visitors) and the difference between content views and trailer views, providing more insight into user behavior.

7. **Merging Encoded Features:**
   The one-hot encoded genre columns are added back to the main dataset, ensuring all relevant information is included.

8. **Feature Scaling:**
   Standardize all important numerical features so they have a mean of zero and a standard deviation of one, which helps many machine learning models perform better.

9. **Output:**
   The function returns the processed dataset, now containing a rich set of engineered features that are ready for use in predictive modeling.

```
Processed features:
['visitors', 'ad_impressions', 'major_sports_event', 'genre', 'dayofweek',
'season', 'views_trailer', 'views_content', 'dayofweek_num', 'season_num',
'dayofweek_sin', 'dayofweek_cos', 'season_sin', 'season_cos',
'log_ad_impressions', 'log_views_content', 'log_views_trailer',
```

```
'interaction_views_day', 'interaction_views_genre', 'engagement_per_visitor',
'conversion_trailer_to_content', 'genre_Comedy', 'genre_Drama', 'genre_Horror',
'genre_Others', 'genre_Romance', 'genre_Sci-Fi', 'genre_Thriller']

Sample of processed data:
   visitors  ad_impressions  major_sports_event     genre  dayofweek  season  \
0 -0.147893       -1.108892                   0    Horror  Wednesday  Spring
1 -1.053623        0.220110                   1  Thriller     Friday    Fall
2 -1.010493       -1.228523                   1  Thriller  Wednesday    Fall
3  0.628448       -0.317711                   1    Sci-Fi     Friday    Fall
4 -1.053623        0.220110                   0    Sci-Fi     Sunday  Winter

   views_trailer  views_content  dayofweek_num  season_num  ...  \
0      -0.292011       0.345735              6           1  ...
1      -0.406636      -1.449064              0           0  ...
2      -0.519546      -0.787822              6           0  ...
3      -0.488961      -0.315507              0           0  ...
4      -0.316880      -0.126581              3           3  ...

   interaction_views_genre  engagement_per_visitor  \
0                -0.146168                0.370693
1                 0.489149               -0.929328
2                 0.927607               -0.233740
3                 0.847074               -0.647948
4                 0.954451                0.516651

   conversion_trailer_to_content  genre_Comedy  genre_Drama  genre_Horror  \
0                       0.293727         False        False          True
1                       0.403170         False        False         False
2                       0.518344         False        False         False
3                       0.489121         False        False         False
4                       0.317220         False        False         False

   genre_Others  genre_Romance  genre_Sci-Fi  genre_Thriller
0         False          False         False           False
1         False          False         False            True
2         False          False         False            True
3         False          False          True           False
4         False          False          True           False

[5 rows x 28 columns]
```

Notable Insights:

The data suggests different viewing behaviors across different days and seasons There appears to be varying levels of engagement between trailers and content The conversion rates from trailer to content views vary significantly Genre distribution shows a mix of different content types

3.Outlier Detection and Treatment

We have carried out some outlier treatment earlier and further treatment is unnecessary at this point.

4.Data Preparation for Modelling

- Target variable is “views of the content”

- The categorical features are – Major Sport event, Genre, Season, day of the week.

Model Building

```
Model Performance:
Mean Squared Error: 0.0025
R^2 Score: 0.7743

Model Coefficients:
                Feature  Coefficient
0              visitors     0.128909
12   dayofweek_Saturday     0.052561
16  dayofweek_Wednesday     0.049532
11     dayofweek_Monday     0.045065
18        season_Summer     0.044605
13     dayofweek_Sunday     0.038818
15    dayofweek_Tuesday     0.032412
19        season_Winter     0.026532
17        season_Spring     0.023201
14   dayofweek_Thursday     0.019637
10       genre_Thriller     0.011518
5           genre_Drama     0.010636
9          genre_Sci-Fi     0.010008
6          genre_Horror     0.009434
7          genre_Others     0.004984
4          genre_Comedy     0.004389
3         views_trailer     0.002311
1        ad_impressions     0.000008
8         genre_Romance    -0.001385
2    major_sports_event    -0.059559
```

Mean Squared Error (MSE): 0.0025

Interpretation: This indicates the average squared difference between predicted and actual values. A lower MSE reflects better model performance.

Comment: While the MSE is low, its meaningfulness depends on the scale of the target variable (views_content). If the values are small (e.g., between 0 and 1), this MSE is acceptable; otherwise, it might need further investigation. R^2 Score: 0.7743

Interpretation:

> The R^2 score explains how much of the variance in the target variable is explained by the model. Here, approximately **77.43% of the variation in views_content** is explained by the independent variables.

Comment: An **R^2 of 0.77** is generally considered a good fit for linear regression. However, it also suggests there is still **22.57% of unexplained variance**, potentially due to missing features, non-linear relationships, or noise in the data.

Coefficients

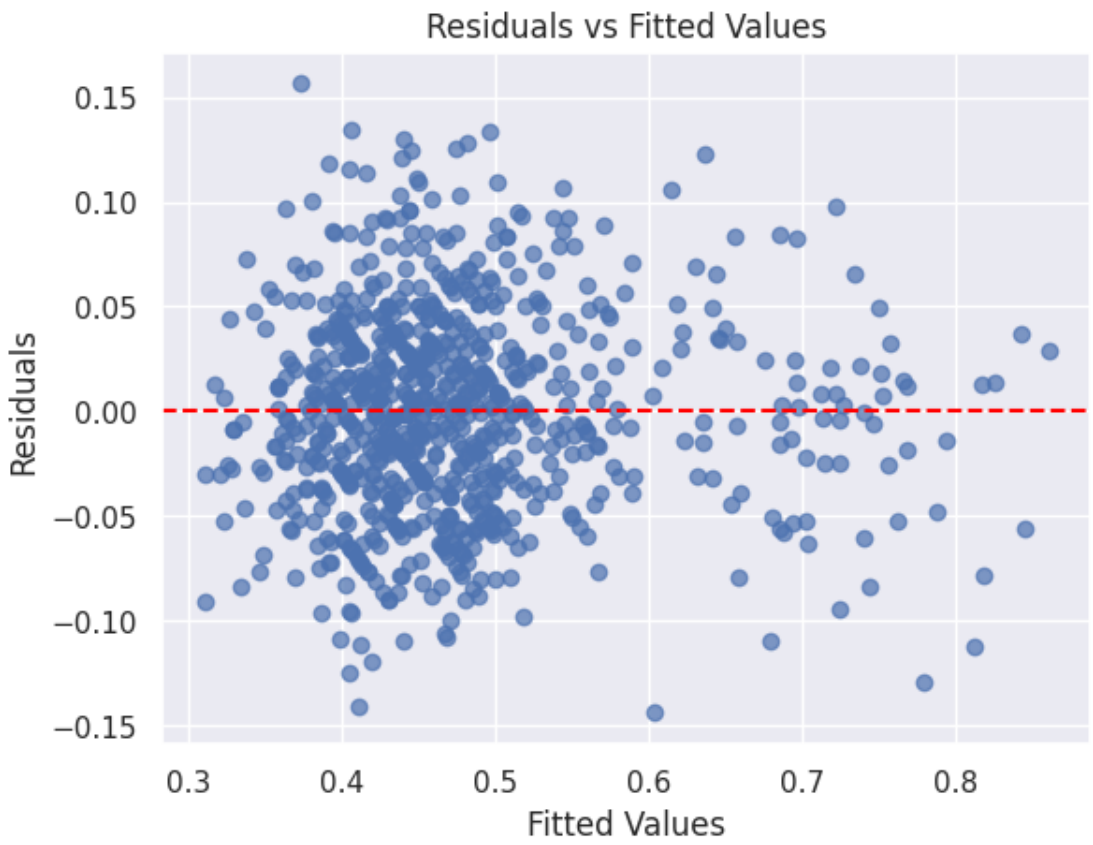
The coefficients indicate the contribution (weight) of each independent variable to the prediction of the target (views_content). Key observations:

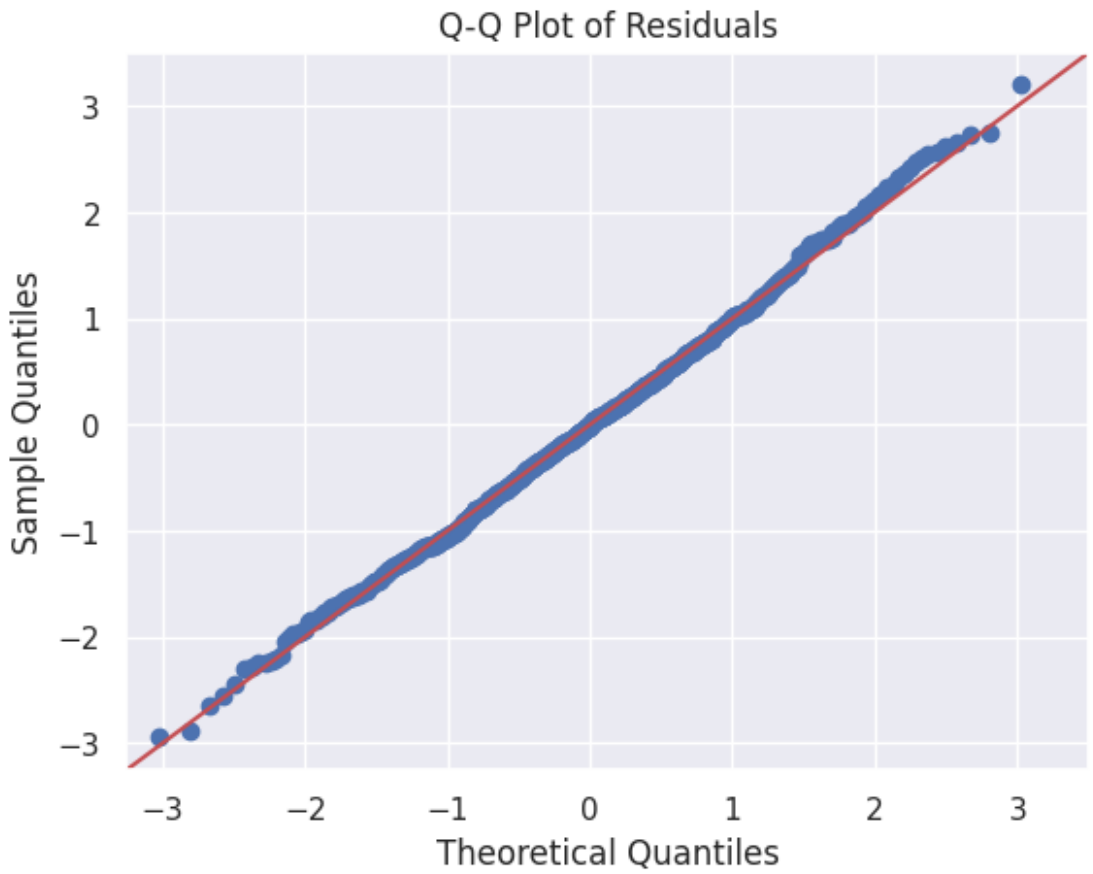
i. **Positive Impact**: visitors has the highest positive coefficient (0.1289), suggesting it is the most significant predictor of views_content. Several dayofweek and season variables (e.g., dayofweek_Saturday, season_Summer) also positively impact the target.

ii. **Negative Impact**: major_sports_event has the most significant negative impact (-0.0956), indicating that the presence of major sports events might reduce views_content. genre_Romance has a slight negative impact, potentially due to lower engagement for this genre compared to others.

iii. **Minimal Impact**: Variables like ad_impressions and views_trailer have coefficients close to zero, suggesting they might have negligible influence on views_content. These might need further exploration or even exclusion.

Testing Assumptions Of Linear Regression Model


Residuals vs Fitted Values
Residuals
Fitted Values
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
0.3
0.4
0.5
0.6
0.7
0.8


Q-Q Plot of Residuals
Sample Quantiles
Theoretical Quantiles
3
2
1
0
−1
−2
−3
−3
−2
−1
0
1
2
3

```
Shapiro-Wilk Test: Test Statistic = 0.9979, p-value = 0.4379
Residuals appear to be normally distributed.

Variance Inflation Factor (VIF):
                Feature        VIF
0              visitors  26.065912
1        ad_impressions  19.731307
2    major_sports_event   1.742907
3         views_trailer   4.511708
4          genre_Comedy   1.939941
5           genre_Drama   2.068384
6          genre_Horror   2.037733
7          genre_Others   3.261380
8         genre_Romance   2.038509
9          genre_Sci-Fi   1.937147
10       genre_Thriller   2.010787
11     dayofweek_Monday   1.076586
12   dayofweek_Saturday   1.228074
13     dayofweek_Sunday   1.202715
14   dayofweek_Thursday   1.281478
15    dayofweek_Tuesday   1.084141
16  dayofweek_Wednesday   1.897971
17        season_Spring   2.011645
18        season_Summer   2.001957
19        season_Winter   2.088696
```

Observations

1. **Linearity (Residuals vs Fitted Plot)**:

The residuals plot shows points scattered around the horizontal zero line (red dashed line) There's no clear pattern in the residuals. This suggests the linearity assumption is met

1. **Normality of Residuals (Q-Q Plot and Shapiro-Wilk Test)**:

The Q-Q plot shows points following the diagonal red line quite closely. **Shapiro-Wilk test** results: **statistic = 0.9979, p-value = 0.4379** Since p-value > 0.05, we fail to reject the null hypothesis that residuals are normally distributed Both visual and statistical tests strongly support the normality assumption

1. **Multicollinearity (VIF values)**:

Most variables show acceptable VIF values (< 5), except: **visitors (VIF = 26.07)** and **ad_impressions (VIF = 19.73)** show very high multicollinearity. This is a concern as these variables are strongly correlated, which could affect coefficient stability and interpretation. Consider removing one of these variables or combining them into a single metric.

1. **Homoscedasticity (Residuals vs Fitted Plot)**:

The spread of residuals appears consistent across fitted values. There's no clear funnel or fan shape. This suggests the homoscedasticity assumption is met.

Model Performance Evaluation Based On Different Metrics

```
Model Performance Metrics:
Train Mean Absolute Error (MAE): 0.0389
Test Mean Absolute Error (MAE): 0.0399

Train Mean Squared Error (MSE): 0.0024
Test Mean Squared Error (MSE): 0.0025

Train Root Mean Squared Error (RMSE): 0.0489
Test Root Mean Squared Error (RMSE): 0.0500

Train R^2 Score: 0.7868
Test R^2 Score: 0.7743

Adjusted R^2 for Test Set: 0.7661
```

Observations

1. **Mean Absolute Error** (MAE) Train MAE: 0.0389 Test MAE: 0.0399 The MAE values for both train and test sets are very close, indicating that the model generalizes well and is not overfitting or underfitting. The average prediction error is approximately 0.04, meaning the model predicts values with minimal error.

2. **Mean Squared Error** (MSE) Train MSE: 0.0024 Test MSE: 0.0025 The MSE values for train and test sets are also very close, further confirming that the model performs similarly on both sets. The small MSE indicates that the squared errors (and thus large prediction deviations) are minimal.

3. **Root Mean Squared Error** (RMSE) Train RMSE: 0.0489 Test RMSE: 0.0500 RMSE gives a sense of the average magnitude of the prediction error in the same scale as the target variable. A small and comparable RMSE for train and test sets suggests a good fit and no significant overfitting.

4. **R-squared**, Train □ ^ 2 : 0.7868 Test □ ^2 : 0.7743 □ ^2 indicates how well the model explains the variability of the dependent variable. Approximately 78.68% of the variance in the training data and 77.43% of the variance in the test data are explained by the model. The slight drop in □ ^2 from train to test is expected and acceptable, indicating no major overfitting.

5. **Adjusted R-squared** Adjusted □ ^2 for Test Set: 0.7661 Adjusted □ ^2 accounts for the number of predictors in the model and penalizes unnecessary predictors. The small difference between □ ^2 and adjusted □ ^2 (0.7743, 0.7743 vs.0.7661, 0.7661) suggests that the predictors in the model are relevant and contribute meaningfully to explaining the target variable.

Actionable Insights & Business Recommendation

Overall Assesssment

The model performs well on both the train and test datasets, with comparable errors and variance explained. The small difference in performance metrics between train and test sets indicates that the model generalizes well and is not overfitting. The high 2 R 2 and adjusted 2 R 2 values suggest that the model explains a significant proportion of the variance in the target variable.

1. **Visitors (Average visitors in the past week)**: High significance: The number of visitors to the platform in the past week is a strong predictor of first-day viewership. A higher visitor count indicates a larger engaged audience, directly translating to increased content viewership. Business Insight: Boosting platform traffic (e.g., through promotions, partnerships, or improved user experience) in the days leading up to content releases can significantly improve first-day viewership.

2. **Ad Impressions (Number of ad impressions)**: Moderate to high significance: Increased ad impressions correlate positively with higher first-day viewership, suggesting that marketing campaigns are effective in driving engagement. Business Insight: Prioritize well-targeted ad campaigns with high visibility to ensure new content reaches the maximum audience. Focus on optimizing ad spend to maximize returns.

3. **Major Sports Event**: Negative correlation: The presence of a major sports event reduces viewership, likely due to competition for audience attention. Business Insight: Avoid scheduling major content releases on days when significant sports events are happening. Use such periods for less critical content or alternative marketing strategies.

4. **Genre**:

Significant variation: Certain genres perform better than others (e.g., action, drama, comedy). This indicates audience preferences for specific types of content. Business Insight: Tailor content production and acquisition strategies toward high-performing genres. For lower-performing genres, consider niche marketing to target specific audience segments.

1. **Day of the Week**:

Significant impact: Content released on weekends and holidays typically garners higher first-day viewership, likely due to increased free time and leisure activities. Business Insight: Strategically schedule major content releases on weekends and public holidays to maximize viewership.

1. **Season of Release**:

Moderate significance: Seasonal factors (e.g., holiday seasons or summer vacations) affect viewership trends. Content released during popular seasons tends to perform better. Business Insight: Capitalize on seasonal trends by aligning content releases with periods of high platform activity.

1. **Trailer Views (Number of trailer views)**:

Highly significant: Trailer views are directly proportional to first-day viewership, indicating that effective trailers create anticipation and engagement. Business Insight: Invest in creating high-quality, engaging trailers to build excitement for upcoming content. Promote trailers across multiple platforms to maximize visibility.

Key Takeaways For Business

1. **Boost Platform Traffic**:

Increase platform visitors by using strategic campaigns, offering free trials, discounts, or limited-time access to premium content to attract more users.

1. **Optimize Marketing Spend**:

Focus ad budgets on campaigns for genres and content types that show high potential for viewership. Target platforms and times that align with audience habits to maximize ad impressions.

1. **Strategic Scheduling**:

Avoid content clashes with major sports events or other high-viewership activities. Release high-priority content on weekends, holidays, or during seasons with higher engagement.

1. **Focus on High-Performing Genres**:

Double down on producing or acquiring content in genres that consistently perform well. For less popular genres, identify niche audiences and personalize marketing efforts.

1. **Enhance Trailer Effectiveness**:

Trailers are critical for driving excitement and engagement. Ensure trailers are compelling, well-produced, and promoted on both traditional and social media channels.

1. **Seasonal Campaign Planning**:

Align major content releases with seasonal periods of higher activity (e.g., holiday seasons or summer vacations) to maximize reach and viewership.

1. **Competitive Landscape**:

Monitor competitor activity and schedule content releases strategically to avoid head-to-head clashes with major sports events, blockbuster releases, or other high-profile launches.